Tianhu Peng

Robotics Researcher | Humanoid Control & Robot Learning

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Profile

PhD researcher at UCL specialising in **humanoid control, robot learning, and real-robot deployment**. Experience spans model-based whole-body control at Tencent Robotics X, reinforcement and imitation learning for locomotion, and streaming generative motion on Unitree humanoids.

Core Skills

Software: Python, C++, PyTorch, ROS/ROS2, DDS, Git, Linux.
Learning: PPO, imitation learning, adversarial motion priors, curriculum learning, diffusion policies, teacher–student distillation.
Control and deployment: whole-body control, task-space tracking, contact constraints, sim-to-real transfer, real-robot integration and debugging.
Tools and platforms: Isaac Gym, Isaac Lab, MuJoCo, Pinocchio, Unitree SDK; Unitree G1/A1.

Industrial Research Experience

Tencent Robotics X Laboratory Jul 2022 – Apr 2023
Robotics Researcher, Control Center Department Shenzhen, China

- Designed and implemented model-based whole-body controllers for bipedal robots.
- Developed task-space objectives, contact constraints, force distribution, and hierarchical task priorities for coordinated motion.
- Integrated controllers with physical robot hardware and validated balance, walking, and whole-body motion execution.

Selected Research and Engineering Projects

AHEADance: Streaming Humanoid Dance Unitree G1

- Developed a robot-native dance generator that anticipates upcoming music using reliability-weighted forecasts of streaming audio.
- Trained continuation from committed motion chunks and deployed generated motion on a physical G1 through whole-body tracking. First-author T-RL submission.

Unified Humanoid Locomotion Unitree G1

- Built a recurrent, gait-conditioned policy with a multi-phase curriculum and reward routing for standing, walking, running, and gait transitions.
- Demonstrated stable standing, walking, and gait transitions on a physical G1. Published at IEEE-RAS Humanoids 2025.

Bipedal Walking with Adversarial Motion Priors Unitree A1

- Combined adversarial motion priors with teacher–student learning; evaluated bipedal walking on flat and challenging terrains in simulation. **TAROS 2024 Best Paper Award**.

Selected Publications and Manuscripts

Complete publication list: Google Scholar.

[1] **T. Peng**, L. Bao, and C. Zhou.
Gait-Conditioned Reinforcement Learning with Multi-Phase Curriculum for Humanoid Locomotion.
IEEE-RAS 24th International Conference on Humanoid Robots (Humanoids), pp. 1087–1093, 2025.

[2] **T. Peng**, L. Bao, J. Humphreys, A. M. Delfaki, D. Kanoulas, and C. Zhou.
Learning Bipedal Walking on a Quadruped Robot via Adversarial Motion Priors.
Towards Autonomous Robotic Systems (TAROS 2024), LNCS 15052, pp. 118–129, 2024. **Best Paper Award**.

[3] **T. Peng**, Y. Wang, L. Bao, Y. Jia, and C. Zhou.
AHEADance: Anticipatory Humanoid Dance from Streaming Audio.
Submitted to IEEE Transactions on Robot Learning (T-RL), 2026; arXiv:2611.13370. Project page.

[4] M. Carroll, **T. Peng**, L. Bao, C. Zhou, and Z. Li.
SCDP: Learning Humanoid Locomotion from Partial Observations via Mixed-Observation Distillation.
arXiv preprint arXiv:2603.09574, 2026.

Education

University College London (UCL) — Nov 2024 – Present
PhD in Robotics – ongoing — London, UK

University of Leeds — May 2023 – Oct 2024
Doctoral Research in Robotics; transferred to UCL — Leeds, UK
Completed first-year doctoral training and transfer viva before continuing at UCL.

Imperial College London — Oct 2020 – Oct 2021
MSc Human and Biological Robotics – Merit — London, UK

University of Sheffield — Sep 2017 – Jul 2020
BEng Mechatronics and Robotics Engineering – First Class Honours — Sheffield, UK

Teaching and Communication

- **UCL, Teaching Assistant – Legged Robot Systems (2025–present):** support practical teaching, coursework, and debugging in robot control, simulation, and reinforcement learning.
- **Imperial College London, Graduate Teaching Assistant – Robotics (2021):** assisted with programming exercises and laboratory work.
- Invited presentation at NVIDIA Robotics Day on humanoid locomotion and distillation-based training in Isaac Lab.

Robot System Deployment

ATEC Humanoid Robot Track: deployed a full-scale Unitree G1 under competition constraints for dynamic balance, outdoor traversal, and teleoperated whole-body manipulation.

Languages

Chinese – Native • English – Fluent

Last updated: 11 November 2026